Aarush Ram Anandh

aarush.anandh_ug2024@ashoka.edu.in | +91 9659786783
github.com/MushuRam01 | linkedin.com/in/aarush-ram-anandh

11 November 2026

Hiring Team
Sony Research India

Application: LLM Research Internship

Dear Hiring Team,

I am applying for Sony Research India's LLM research internship. I am completing a B.Sc. (Hons.) in Computer Science and Mathematics at Ashoka University, expected in 2028. Your emphasis on understanding model behaviour and developing methods through hypotheses and experiments is particularly appealing to me. My hands-on work is in diffusion models, reinforcement learning, audio classification and Transformer-based financial modelling.

At the Constrained Image-Synthesis Lab, I investigated inference-time colour preservation in latent diffusion, achieving $\Delta E_{00} < 1$ and approximately four times better colour accuracy than evaluated open-source baselines. The work is reported in my public preprint, *Inference-Time Loss-Guided Colour Preservation in Diffusion Sampling* (arXiv:2601.17259). I am also developing a controllable optimal-transport framework, CGSW, with a paper submitted to JMLR. These projects have involved studying why a method behaves as it does, translating mathematical ideas into code and choosing experiments that test the intended claim.

In an ongoing two-person RL project, we co-developed masked-action PPO and DQN agents and evaluated PPO across two team variants and five seeds each. PPO reached 99.7–100% wins against the fixed heuristic over 1,020 final battles per run; confidence intervals and permutation tests showed no significant learning-speed difference between variants. My MEIKURAL project classifies seven Hindustani raags with STFT/CNN representations trained on 3,471 audio windows; at the Koita Centre for Digital Health, I contributed to a wearable-data pipeline achieving approximately 70% sensitivity for asthma exacerbation detection in study evaluation. I also built experimental frozen time-series foundation-model routes for surprise features and embeddings, implemented but not yet run in the reported cohort evaluation.

At Avartan Labs, I raised rare-class correct detection from 40% to 80% in an RT-DETR pipeline covering 18 material classes. This industry internship gave me experience analysing detector failures and investigating targeted model improvements.

I would welcome the opportunity to apply this experimental approach to multilingual model behaviour, reasoning and entertainment-domain adaptation. I can contribute Python/PyTorch prototyping and careful analysis while developing the language-model-specific expertise the work requires. Thank you for considering my application.

Yours sincerely,

Aarush Ram Anandh